

KUMAR SAMYAK

Computer Science Student | Robotics, Embedded Systems & Computer Vision
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EDUCATION

University of Illinois Urbana-Champaign | Urbana-Champaign, IL

Expected May 2030

Bachelor of Science in Computer Science | Intended Minor in Electrical Engineering

Relevant Coursework: Introduction to Computer Science II (C++); Discrete Structures

Technical Organizations: Illini Electric Motorsports, Illinois Robotics in Space, SIGRobotics, Illini Pullers, Liquid Rocketry

Independence High School | Frisco, TX

August 2022 - May 2026

Weighted GPA: 5.335/6 | Top 4.9% | SAT: 1540

Awards and Certifications:

- Space Center University at NASA: 1st Place, Internal Robotics Competition (2023)
- Harvard CS50: CS50x Intro to CS, CS50P Intro to Programming with Python, CS50AI Intro to AI with Python

TECHNICAL EXPERIENCE & LEADERSHIP

Knights Robotics - FRC Team 6901

Aug 2022 - May 2026 | [Website](#)

Co-President & Programming Lead

Frisco, TX

- Developed Java robot codebases using WPILib, PhotonVision, PathPlanner, and CTRE Phoenix 6 for field-centric swerve drive, odometry, PID control, and vision-based autonomous routines; led Fusion 360-based CNC machining and 3D printing of custom robot components.
- Recruited and managed approximately 50 students, mentored new programmers through development of the 2026 codebase, and coordinated design, manufacturing, mechanical, electrical, and programming subteams.
- Led the team to earn two Texas State Championship qualifications, a second-place District Championship finish, District Finalist recognition, and Sustainability, Rising All-Star, and two Gracious Professionalism awards.

Robotics Internship

June 2025

PID Line-Following Robot

UT Austin, TX

- Soldered and programmed a C++ line-following robot, tuning motors, calibrating sensors, and optimizing PID control to win first place with the fastest internal competition time.

AI/ML Internship

June 2024 - July 2024 | [GitHub](#)

Handwritten Character Recognition

UT Dallas, TX

- Built and evaluated a convolutional neural network in Python and TensorFlow to classify handwritten letters and digits, then presented the project to professors and peers at the final showcase.

Academic Empowerment Project Inc.

Aug 2022 - May 2024

Vice President of Technology

Frisco, TX

- Developed a class-management Discord bot in Python using the Discord and Google Calendar APIs, enabling teachers to organize courses, manage registrations, distribute syllabi, and coordinate class schedules.
- Taught foundational Python to five elementary and middle school students and developed math and computer science curricula supporting approximately 100 students nationwide.

PROJECTS

Gym Bro - Computer-Vision Workout Assistant | Personal Project

Aug 2026 - Present | [GitHub](#)

- Built a browser-based workout assistant with JavaScript, Vite, MediaPipe Pose Landmarker, getUserMedia, and the Canvas API for real-time local pose tracking.
- Implemented side-view bicep-curl counting with automatic visible-arm selection, elbow-angle calculation, confidence filtering, smoothing, and a movement state machine.
- Actively expanding the application to support multiple exercises, automatic set tracking, real-time form feedback, and progress analytics using exercise-specific pose and joint-angle data.

Five-Finger Tendon-Driven Robotic Hand | Independent Study and Research

Aug 2024 - May 2026 | [Portfolio](#)

- Designed and fabricated a fully 3D-printable tendon-driven hand in Fusion 360 with embedded MG90S servos, custom tendon spools, TPU return springs, and palm-mounted electronics.
- Designed the control circuit around Raspberry Pi hardware and a PCA9685 PWM driver; developed Python for autonomous grasping and individual finger control for achieving various grasps and poses.
- Iterated from an unsuccessful four-bar finger mechanism to tendon drive; experimentally tested 95A and 85A TPU return elements to balance passive extension force against servo torque.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, JavaScript, HTML/CSS

Libraries & Robotics: TensorFlow, OpenCV, MediaPipe, WPILib, PhotonVision, PathPlanner, CTRE Phoenix 6

Tools: Git/GitHub, VS Code, Vite, Arduino IDE, Autodesk Fusion 360, Multisim

Hardware & Fabrication: Raspberry Pi 3 B+/Pico, Orange Pi, CAN, 3D printing, CNC machining, soldering